

Oxytocin: Drug information - UpToDate

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For additional information see "Oxytocin: Patient drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Appropriate use:

Elective induction of labor is defined as the initiation of labor in a pregnant individual who has no medical indications for induction. Since the available data are inadequate to evaluate the benefits-to-risks considerations, oxytocin is not indicated for elective induction of labor.

Brand Names: US

- Pitocin

Pharmacologic Category

- Oxytocic Agent

Dosing: Adult

Labor induction or augmentation

Labor induction or augmentation: Note: To minimize medication errors, administration of the dose using a standardized concentration and infusion protocol is recommended (Ref). Administration requires continuous fetal and uterine monitoring. The ideal dosing regimen has not been established and various protocols are available (Ref); examples of regimens are listed below. Refer to institution-specific protocol.

Low-dose regimen (off-label dose): IV: Initial: 0.5 to 2 milliunits/minute; incrementally increase by 1 to 2 milliunits/minute every 15 to 40 minutes as needed according to response (eg, contractions and fetal heart rate) (Ref). Reduce dose or discontinue in the event of uterine tachysystole and/or fetal distress. See "Dose adjustment for oxytocin-induced uterine tachysystole."

High-dose regimen (off-label dose): IV: Initial: 4 to 6 milliunits/minute; incrementally increase by 3 to 6 milliunits/minute every 15 to 40 minutes as needed according to response (eg, contractions and fetal heart rate) (Ref). Reduce dose or discontinue in the event of uterine tachysystole and/or fetal distress. See "Dose adjustment for oxytocin-induced uterine tachysystole."

Maximum dose: Not clearly established. Many protocols suggest a maximum dose of 40 milliunits/minute; however, higher doses (eg, 90 milliunits/minute) have been used (Ref). In patients with prior cesarean birth, lower maximum doses (eg, 20 milliunits/minute) may be preferred (Ref).

Dose adjustment for oxytocin-induced uterine tachysystole:

Normal fetal heart rate: Reduce the dose (eg, decrease oxytocin infusion rate by 50%) or discontinue oxytocin until tachysystole resolves (Ref). Practice varies and depends on multiple factors (eg, frequency of contractions, dose of oxytocin when tachysystole occurred).

Indeterminate or abnormal fetal heart rate: Discontinue oxytocin and initiate other measures (eg, maternal repositioning, administration of IV fluids) (Ref). Prompt administration of another agent (eg, terbutaline) may be needed if fetal heart rate (FHR) changes are concerning for acidemia (Ref).

Restarting infusion: Following resolution of uterine tachysystole when contraction frequency, intensity, and duration and FHR are normal, oxytocin may be restarted (eg, if discontinued for <30 minutes, restart oxytocin at no more than 50% of the rate that caused the tachysystole and gradually adjust dose based on protocol and maternal/fetal status; if discontinued for >30 minutes, restart at the initial infusion rate) (Ref).

Discontinuation of therapy in the active phase of labor: May continue infusion through delivery or discontinue infusion once cervical dilation is 5 to 6 cm and adequate contractions (Ref). **Note:** If discontinued, oxytocin is typically restarted after delivery of the placenta, delivery of the anterior shoulder, or after delayed umbilical cord clamping for prevention of postpartum uterine hemorrhage (Ref). Refer to "Postpartum uterine hemorrhage."

Medication abortion, second trimester

Medication abortion, second trimester (alternative agent) (off-label use): **Note:** Alternative if misoprostol is not available.

IV: Initial: 20 to 100 units over 3 hours, then stop the infusion for 1 hour to allow for diuresis. Restart the infusion by slowly increasing the rate at 3-hour intervals (eg, by 50 units every 3 hours) until fetal expulsion occurs; maximum rate: 300 units over 3 hours (Ref).

Postpartum uterine hemorrhage

Postpartum uterine hemorrhage: **Note:** To minimize medication errors, administration of the dose using a standardized concentration and infusion protocol is recommended (Ref). The optimal IV regimen has not been established (Ref); examples of regimens are listed below. Refer to institution-specific protocol.

Prevention of postpartum hemorrhage: **Note:** Administer oxytocin after delivery of the placenta, delivery of the anterior shoulder, or after delayed umbilical cord clamping (Ref). In patients at high risk for hemorrhage, a second medication (eg, tranexamic acid, misoprostol) may be used; some experts use combination therapy in all patients regardless of hemorrhage risk (Ref).

IV: 10 units over 30 minutes, then decrease rate to 7.5 units/hour. Alternatively, may administer 1 to 3 units as a **slow** IV bolus followed by a maintenance infusion of up to 7.5 units/hour (Ref). Adjust infusion rate to sustain uterine contraction and control uterine atony, and then decrease rate (eg, to 1 to 2.5 units/hour) if uterine tone is maintained and bleeding is not excessive (Ref). Usual duration is 2 to 4 hours, but longer durations (up to 8 hours) may be used in patients at high risk for hemorrhage (Ref). If previously administered during labor, larger infusion doses may be necessary (Ref).

IM: 10 units given once (Ref).

Treatment of postpartum hemorrhage: **Note:** Prompt administration of uterotonic medication(s) for postpartum hemorrhage (an obstetric emergency) is required; if hemorrhage is not controlled by medications, minimally invasive measures and/or emergency invasive intervention may also be warranted. The optimal IV regimen has not been established; an example regimen is listed below. Refer to institution-specific protocol.

IV: 30 to 40 units/hour for 10 to 15 minutes to control uterine atony, followed by a continuous lower rate to maintain uterine contractions (Ref). If already administered IV for preventive therapy (most patients), may increase infusion rate (Ref). If previously administered during labor, larger infusion doses may be necessary (Ref).

IM: 10 units given once in conjunction with other interventions (Ref). **Note:** If already administered IM for preventative therapy, treatment is typically not repeated (given the long duration of action with IM use). Intramyometrial administration may also be used (Ref).

Maximum dose: Not clearly established; some guidelines recommend a maximum cumulative dose of 40 units for both prevention and treatment (Ref). Some data suggest a greater risk of cardiovascular adverse effects with high doses over a shorter period of time or rapid IV boluses than with higher cumulative doses over a longer period of time (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely as limited excretion of active drug in the urine.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Cardiac arrhythmia, hypertensive crisis, hypotension (Dyer 2011), subarachnoid hemorrhage, tachycardia (Dyer 2011), ventricular premature contractions

Endocrine & metabolic: Water intoxication (severe water intoxication with seizure and coma is associated with a slow oxytocin infusion over 24 hours)

Gastrointestinal: Nausea, vomiting

Genitourinary: Postpartum hemorrhage, uterine rupture

Hematologic & oncologic: Pelvic hematoma

Hypersensitivity: Anaphylaxis

Contraindications

Hypersensitivity to oxytocin or any component of the formulation; significant cephalopelvic disproportion; unfavorable fetal positions or presentations (such as transverse lies); fetal distress when delivery is not imminent; hypertonic or hyperactive uterus; contraindicated vaginal delivery (invasive cervical cancer, active genital herpes, prolapse of the cord, cord presentation, total placenta previa, or vasa previa); obstetrical emergencies where surgical intervention is favored; where adequate uterine activity fails to achieve satisfactory progress.

Canadian labeling: Additional contraindications (not in the US labeling): Severe toxemia; prematurity or unripe cervix; predisposition to uterine rupture (eg, grand multiparity, overdistention of the uterus, previous caesarian delivery, other surgery involving the uterus); prolonged use in uterine inertia; factors predisposing to thromboplastin or amniotic fluid embolism (eg, prolonged retention of dead fetus, placental abruption); serious medical or obstetric conditions and any condition in which fetal distress already occurs; inability of physician to be in attendance

Warnings/Precautions

Concerns related to adverse effects:

- Antidiuretic effect: Oxytocin may produce intrinsic antidiuretic effect (ie, water intoxication). Severe water intoxication with convulsions, coma, and death may occur, particularly with large doses (40 to 50 milliunits/minute) administered as a slow infusion over 24 hours and if the patient is receiving additional fluids.

- Cardiovascular effects: Arrhythmias, hypotension, myocardial ischemia, peripheral vasodilation, and tachycardia have been reported following administration. The risk of adverse events is influenced by dose and route of administration and is increased in patients with cardiovascular disease. Use with extreme caution in hemodynamically unstable patients (Dyer 2011).
- Maternal deaths: Maternal deaths caused by hypertensive episodes, subarachnoid hemorrhage, or rupture of the uterus and fetal deaths have occurred with oxytocic medications when used for induction of labor or for augmentation in the first and second stages of labor.
- Uterine effects: High doses or hypersensitivity to oxytocin may cause uterine hypertonicity, spasm, tetanic contraction, or rupture of the uterus. Prolonged infusion of oxytocin may saturate uterine receptors, causing inadequate uterine contractions and prolonged labor (AWHONN [Simpson 2020], Valleria 2017).

Other warnings/precautions:

- Appropriate use: Labor induction: Oxytocin is used to initiate or improve uterine contractions in order to replicate spontaneous labor and achieve a vaginal delivery. Use for induction of labor is generally not recommended in the following conditions: Fetal distress, hydramnios, partial placenta previa, prematurity, borderline cephalopelvic disproportion, or conditions where there is a predisposition for uterine rupture (eg, previous major surgery on cervix or uterus, cesarean section, overdistention of the uterus, grand multiparity, past history of uterine sepsis or traumatic delivery).
- Trained personnel: IV preparations should be administered by adequately trained individuals familiar with its use and able to identify complications; continuous observation is necessary for all patients. **The use of one standardized concentration for the induction or augmentation of labor and the management of postpartum uterine bleeding is recommended to minimize errors** (AWHONN 2021; AWHONN [Simpson 2020]; ISMP [Smetzer 2022]). Administration of ready-to-use bags boldly labeled on both sides (to differentiate from other solutions), not bringing the prepared oxytocin infusion to the patient's bedside until needed, and the use of standardized order sets are additional best practice safety interventions (ISMP [Smetzer 2022]).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Pitocin: 10 units/mL (1 mL, 10 mL, 50 mL) [contains chlorobutanol (chlorobutol)]

Generic: 10 units/mL (1 mL, 10 mL, 30 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Oxytocin Injection)

10 units/mL (per mL): \$1.80 - \$4.14

Solution (Pitocin Injection)

10 units/mL (per mL): \$1.68

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect,

incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Generic: 10 units/mL (1 mL, 5 mL, 10 mL)

Administration: Adult

The use of one standardized concentration and administration protocol (eg, the rate and interval for dose increases) for the induction or augmentation of labor and the management of postpartum uterine bleeding is recommended to minimize errors (Ref). Administration of ready-to-use bags boldly labeled on both sides (to differentiate from other solutions), not bringing the prepared oxytocin infusion to the patient's bedside until needed, and the use of standardized order sets are additional best practice safety interventions (Ref).

Labor induction or augmentation: IV administration requires the use of an infusion pump that allows minute-to-minute adjustments; refer to indication-specific infusion rates in dosing for detailed recommendations. The oxytocin solution can be piggybacked to an isotonic electrolyte infusion line proximal to the venipuncture site (Ref).

Postpartum uterine hemorrhage: Administer by IV or IM. IM administration may be used when IV access is not available (Ref). IV push is not recommended; rapid IV bolus administration is associated with cardiovascular collapse (Ref). Slow IV injections (5 or 10 units over 1 minute) are preferred for patients without cardiovascular risk factors; very slow injections (>5 minutes) are preferred for patients with cardiovascular risk factors (Ref); refer to indication-specific infusion rates in dosing for detailed recommendations. Administration IV is preferred over IM injection for the prevention of postpartum hemorrhage following vaginal delivery in patients who already have IV access (Ref). Either route of administration may be used in patients undergoing cesarean delivery (Ref).

Medication abortion: Administer by IV infusion; refer to indication-specific infusion rates in dosing for detailed recommendations.

Preparation for Administration: Adult

IV: Note: Although there are various options for preparation, **the use of one standardized concentration for the induction or augmentation of labor and the management of postpartum uterine bleeding is recommended to minimize errors** (Ref). Providing oxytocin in ready-to-use bags boldly labeled on both sides (to differentiate from other solutions) is an additional best practice safety intervention (Ref).

Labor induction or augmentation: **Use of a ready-to-use bag of an oxytocin infusion with a standardized concentration is recommended** (refer to institution-specific protocol) (Ref). Oxytocin 30 units added to 500 mL NS or LR provides a concentration of 60 milliunits/mL, allowing infusion pump settings (mL/hour) to match dose administration (milliunits/minute) (Ref). Alternately, oxytocin 10 units may be added to NS or LR 1,000 mL to yield a solution containing oxytocin 10 milliunits/mL (per manufacturer labeling) or 80 units to 1,000 mL D5W, NS, or LR to provide a solution of 80 milliunits/mL (Ref). Rotate solution to mix.

Postpartum uterine hemorrhage: **Use of a ready-to-use bag of an oxytocin infusion with a standardized concentration is recommended** (refer to institution-specific protocol) (Ref). Although there is no standard for preparation, one option is to add oxytocin 30 units to 500 mL NS or LR (Ref).

Medication abortion: Add oxytocin 10 units to 500 mL of NS solution or D5W.

Hazardous Drugs Handling Considerations

Hazardous agent (NIOSH 2024 [table 2]).

Use appropriate precautions for receiving, handling, storage, preparation, dispensing, transporting, administration, and disposal. Follow NIOSH and USP 800 recommendations **and** institution-specific policies/procedures for appropriate containment strategy (NIOSH 2023; NIOSH 2024; USP-NF 2020).

Note: Facilities may perform risk assessment of some hazardous drugs to determine if appropriate for alternative handling and containment strategies (USP-NF 2020). Refer to institution-specific handling policies/procedures.

Storage/Stability

Store intact vials/containers at 20°C to 25°C (68°F to 77°F).

Use: Labeled Indications

Labor induction or augmentation: Antepartum use for the initiation or improvement of uterine contractions, where this is desirable and considered suitable for reasons of fetal or maternal concern, in order to achieve vaginal delivery. Induction of labor in patients with a medical indication (eg, Rh problems [isoimmunization], maternal diabetes, preeclampsia, at or near term) when delivery is in the best interests of mother and fetus or when membranes are prematurely ruptured, and delivery is indicated; stimulation or reinforcement of labor (as in selected cases of uterine inertia).

Note: Oxytocin may also be used for other medical conditions in antepartum labor induction, including chronic or gestational hypertension, fetal compromise (eg, severe fetal growth restriction or oligohydramnios), or augmentation of labor, used when contractions are inadequate to produce a cervical change in dilation and effacement (ACOG 2009; AWHONN [Simpson 2020]).

Medication abortion, second trimester:

Limitations of use: Oxytocin monotherapy may be used as an alternative to misoprostol for second trimester medication abortion if misoprostol is not available (ACOG 2013; Zwerling 2023). **Note:** Dosing in the prescribing information for the adjunctive treatment of abortion may not reflect current clinical practice; other therapy may be required.

Postpartum uterine hemorrhage: To produce uterine contractions during the third stage of labor and to control postpartum hemorrhage.

Use: Off-Label: Adult

Labor induction, elective

Medication Safety Issues

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drugs which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Sound-alike/look-alike issues:

Oxytocin may be confused with OxyCONTIN

Pitocin may be confused with PIT or Pitressin (names to describe vasopressin), pitolisant

Other safety concerns:

OXY is an error-prone abbreviation (mistaken as oxyCODONE, OxyCONTIN)

PIT is an error-prone abbreviation for Pitocin (mistaken as Pitressin [a discontinued brand of vasopressin still referred to as PIT])

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Amisulpride (Oral): May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk C: Monitor*

Azithromycin (Systemic): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Azithromycin (Systemic). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Carbetocin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Carbetocin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Carboprost Tromethamine: May increase adverse/toxic effects of Oxytocic Agents. Specifically, oxytocic effects may be enhanced. Management: Consider alternatives when possible, as product labeling states that the use of carboprost tromethamine in combination with oxytocic agents is not recommended. If combined, the risk for uterine overstimulation may be increased. *Risk D: Consider Therapy Modification*

Chloroquine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Chloroquine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Clofazimine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Clofazimine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Dabrafenib: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Dabrafenib. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Dinoprostone: May increase adverse/toxic effects of Oxytocin. Specifically, oxytocic effects may be enhanced. Management: Concomitant use of dinoprostone and oxytocin is not recommended. If used sequentially, monitor uterine activity closely. Administer oxytocin 30 minutes after removing dinoprostone vaginal insert and 6 to 12 hours after the application of dinoprostone gel. *Risk D: Consider Therapy Modification*

Domperidone: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Domperidone. *Risk X: Avoid*

Encorafenib: May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

EPHEDrine (Nasal): Oxytocin may increase hypertensive effects of EPHEDrine (Nasal). *Risk C: Monitor*

EPHEDrine (Systemic): Oxytocin may increase hypertensive effects of EPHEDrine (Systemic). *Risk C: Monitor*

Fexinidazole: May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these

agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Fluorouracil Products: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Fluorouracil Products. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Gadobenate Dimeglumine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Gadobenate Dimeglumine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Gemeprost: May increase adverse/toxic effects of Oxytocin. Management: Consider alternatives to the coadministration of gemeprost and labor inducers, such as oxytocin, when possible as product labeling states the concomitant use of these agents is not recommended. If used sequentially, monitor uterine activity closely. *Risk D: Consider Therapy Modification*

Halofantrine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Halofantrine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Haloperidol: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Haloperidol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Inotuzumab Ozogamicin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Inotuzumab Ozogamicin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Levoketoconazole: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Levoketoconazole. *Risk X: Avoid*

Lofexidine: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Lofexidine. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Midostaurin: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Midostaurin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

MiSOPROStol: May increase adverse/toxic effects of Oxytocic Agents. Specifically, the oxytocic effects may be increased. Management: The concomitant use of misoprostol with other oxytocic agents is not recommended. If sequential use of oxytocin is necessary, oxytocin should be given at least 4 hours after misoprostol. *Risk D: Consider Therapy Modification*

Ondansetron: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Ondansetron. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pentamidine (Systemic): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Pentamidine (Systemic). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pimozide: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Piperaquine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Piperaquine. *Risk X: Avoid*

Probucol: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Probucol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): Oxytocin may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Antidepressants (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Antipsychotics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Antipsychotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class IC Antiarrhythmics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Class IC Antiarrhythmics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-Prolonging Inhalational Anesthetics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Kinase Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Miscellaneous Agents (Moderate Risk): May increase QTc-prolonging effects of Oxytocin. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk) may increase serum concentrations of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Quinolone Antibiotics (Moderate Risk): QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Quinolone Antibiotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Strong CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-prolonging Miscellaneous Agents (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Sertindole: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Succinylcholine: Oxytocin may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Sulprostone: May increase adverse/toxic effects of Oxytocic Agents. Management: Consider alternatives, as the use of sulprostone in combination with oxytocic agents is not recommended. If combined, the risk for uterine overstimulation may be increased. This does not apply when sulprostone is used for postpartum atonic bleeding *Risk D: Consider Therapy Modification*

Thioridazine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Thioridazine. *Risk X: Avoid*

Toremifene: QT-prolonging Miscellaneous Agents (Moderate Risk) may increase QTc-prolonging effects of Toremifene. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pregnancy Considerations

Small amounts of exogenous oxytocin are expected to reach the fetal circulation. When used as indicated, teratogenic effects would not be expected. Nonteratogenic adverse reactions are reported in the neonate as well as the mother.

Oxytocin is used for augmentation of labor in patients making slow progress through the first stage of spontaneous labor. The time to delivery is shortened by ~2 hours (Alhafez 2020).

When used for induction of labor, maternal and fetal conditions, cervical status, gestational age, and other factors should be considered. Indications for induction of labor are not absolute but may include abruptio placentae, chorioamnionitis, post-term pregnancy, preeclampsia/eclampsia, premature rupture of membranes, fetal compromise (eg, severe fetal growth restriction, isoimmunization, oligohydramnios), fetal demise, or maternal medical conditions (eg, diabetes mellitus, kidney disease, chronic pulmonary disease, chronic or gestational hypertension, antiphospholipid syndrome) (ACOG 2009). Oxytocin should not be used to induce labor when spontaneous labor or vaginal birth would otherwise be contraindicated (ACOG 2009; AWHONN [Simpson 2020]).

Although the prescribing information contains a boxed warning against use for non-medical induction of labor, oxytocin may be used off-label for the elective induction of labor at 39 weeks gestation. Details for this use come from the ARRIVE trial (A Randomized Trial of Induction Versus Expectant Management). The study included low risk nulliparous patients with singleton gestations. Eligible patients did not have conditions requiring delivery prior to 40 weeks gestation and had reliable information of gestational age based on ultrasound and date of last menstrual period. A specific dosing protocol for oxytocin was not used in the study. The study showed no statistical difference in the primary composite outcome of perinatal mortality and severe perinatal morbidity while noting that cesarean delivery rate was significantly lower in the induction of labor group. In addition, there was a lower rate of gestational hypertension and preeclampsia as well as a reduced need for neonatal respiratory support in the first 72 hours of life. The decision to induce labor in full-term nulliparous patients without a medical indication for delivery should be based on a shared decision-making process that compares the benefits and risks of labor induction and expectant management, the patients' birthing preferences, and the resources available (ACOG 2025; Grobman 2018; SMFM 2019).

Significant drug interactions exist (eg, dinoprostone, misoprostol), requiring dose/frequency adjustment or avoidance (ACOG 2009). Consult drug interactions database for more information.

Breastfeeding Considerations

Endogenous oxytocin mediates milk ejection. Administration of exogenous oxytocin may negatively impact breastfeeding. However, available studies have inconsistent results when evaluating the onset of lactogenesis, or the initiation and duration of breastfeeding. Outcomes may be influenced by study design, maternal dose, and indications for use (Buckley 2015; Erickson 2017; Fernández-Cañadas 2017; Fernández-Cañadas 2019; Gomes 2018).

Monitoring Parameters

Fluid intake and output during administration, uterine activity (tonus, amplitude, and frequency of contractions), maternal blood pressure; continuous electronic fetal heart rate monitoring in relation to uterine contractions.

Mechanism of Action

Oxytocin stimulates uterine contractions by acting on receptors that trigger the release of intracellular calcium and local prostaglandin production. Oxytocin specific receptors are not present in the uterus until ~13 weeks' gestation and increase as pregnancy progresses and reach maximum concentration at term. Therefore, term pregnancies are more sensitive to lower oxytocin doses. The action of oxytocin is limited by the concentration of receptors in the smooth muscle of the uterus. Repeated doses may cause desensitization of the receptors and decreased response (Vallera 2017).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Uterine contractions: IM: 3 to 5 minutes; IV: ~1 minute

Duration: IM: 2 to 3 hours; IV: 1 hour

Half-life elimination: 1 to 6 minutes; decreased in late pregnancy and during lactation

Excretion: Urine (small amount unchanged)

Patient Counseling Points

(For additional information see "Oxytocin: Patient drug information")

What is this drug used for?

- It is used to start labor.
- It is used to control bleeding that happens after giving birth.
- It is used to help manage some types of abortion.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Upset stomach or throwing up

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Bleeding a lot (soaking 1 pad per hour)
- Slow heartbeat
- A heartbeat that does not feel normal
- Trouble passing urine
- Severe stomach pain
- Seizures
- Weakness on 1 side of the body, trouble speaking or thinking, change in balance, drooping on one side of the face, or blurred eyesight
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Pitocin | Syntocinon;
- (AR) Argentina: Hipofisina | Veracuril;
- (AU) Australia: Oxytocin apotex;
- (BD) Bangladesh: Ocina | Pitocin;
- (BR) Brazil: Orastina | Syntocinon;
- (CL) Chile: Syntocinon;
- (CO) Colombia: Oxitocina | Pitocin | Syntocinon;
- (CZ) Czech Republic: Ofost | Oxytocin avmc;
- (DO) Dominican Republic: Oxitocina;
- (EC) Ecuador: Oxitocina | Pitocin;
- (EG) Egypt: Sunnilyabcin | Syntocinon;
- (FI) Finland: Partolact medica;
- (GB) United Kingdom: Syntocinon;
- (HR) Croatia: Syntocinon;
- (HU) Hungary: Oxygrindeks;
- (IE) Ireland: Ofost | Syntocinon;
- (IL) Israel: Oxyd | Syntocinon;
- (IN) India: Genox | Gynotocin | Itocin | Oxytin | Oxytomed | Oxyton | Pitocin;
- (IT) Italy: Syntocinon;
- (JP) Japan: Atonin;
- (KE) Kenya: Pepcinon | Tocinox;
- (KR) Korea, Republic of: Jw oxytocin;
- (LT) Lithuania: Ofost | Oxytocin panpharma;
- (MX) Mexico: Syntocinon;
- (NL) Netherlands: Oxygrindeks | Syntocinon;
- (NO) Norway: Oxytocin aurora medical | Oxytocin life | Pitocin;
- (NZ) New Zealand: Oxytocin apotex;
- (PH) Philippines: Ambitocyn | Cleartocin | Fetusin | Lavoxin | Ocitoxin | Oxitex | Oxitone | Oxynox | Solvoxine | Tocinox | Tranox;
- (PK) Pakistan: Syntocinon;
- (PL) Poland: Syntocinon;
- (PT) Portugal: Oxitocina | Oxitocina hikma;
- (PY) Paraguay: Veracuril;
- (QA) Qatar: Pitocin | Syntocinon;
- (RO) Romania: Oxitocina panpharma | Oxitocina s;
- (SA) Saudi Arabia: Gynocin;

- (SE) Sweden: Oxytocin Pilum;
- (SK) Slovakia: Ofost | Oxygrindeks;
- (TH) Thailand: Oxytocin synth;
- (TR) Turkey: Postuitrine | Synpitan | Syntocinon;
- (TW) Taiwan: Litocin | Pitocin | Piton S | Syntocinon;
- (UY) Uruguay: Oxitocina | Veracuril;
- (VE) Venezuela, Bolivarian Republic of: Syntocinon;
- (ZA) South Africa: Oxyject | Spec oxytocin

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